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Autism Screening Using a Video Game Based on Emotions

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***Abstract*— Autism spectrum disorders (ASDs) is referred to as a range of mental and physical. Usually these children deal with disorders such as social problems, repeated patterns of behaviors and inability to understand abstract concepts. If at the very childhood the occupational therapy exercises are done on a person with autism, he can return to normal life. Given the importance of early diagnosis of autism, this study presents the structure of a serious smart game for screening and rehabilitation of autism. This computer game is based on training of three important skills in understanding emotions i.e. the recognition of emotions, the emergence of emotions, and emotion regulation. The game was performed on children with autism and normal children for designing pattern-recognition algorithms for screening children with autism. Then the screening precision reached 93% using a support vector machine with Gaussian kernel.**

Keywords*— *autism, Serious Game, emotion expression, emotion recognition, emotion regulation, autism screening.

I. Introduction

Autism is one of the evolutionary disorders which occur during childhood or early childhood but there has been recent evidence suggesting that autism can be detected in the first months of a child's life [1]. However, the diagnosis of autism in the first months of the first year life is not easy because the children's patterns of behavior are not completely clear before the age of two. Mental retardation may also be considered in cases where the child is also afflicted with other disorders, such as mental retardation, and the child's autism may be ignored. In addition, one of the major problems caused by autism is related to language and speaking, so detection can only take place when this ability is developed in the child. It should be noted that some autistic children first develop relatively naturally and their autism is evident only when they are two years old and begin to lose their previous skills.

Experts usually recognize autism before the child reaches the age of three, but sometimes this diagnosis is delayed for some reasons. The most commonly used methods for the diagnosis of autism are questionnaire-based methods, parental reports [2], [3], medical tests [4] such as brain scans and blood tests, as well as intelligent screening methods.

In intelligent autism screening, is tried to use the differences between normal and people with autism and use artificial intelligence to develop a tool for the screening and diagnosis of autism. Of course, smart devices are not developed enough that enable us to definitely diagnose whether or not there is a disorder without expert comments and only through a screening tool. artificial intelligence in applied in autism screening to use the way eyes look [5], [6], how to walk [7], how to play with a toy car [8], how to play a hand-eye coordination video game [9], and intonation [10].

Recent studies have revealed that children with autism have serious flaws in face recognition skills, while there is no significant difference in the perceptual abilities of these children with others [11]. Furthermore, these children are less sensitive to negative emotions such as sadness and fear [12]. With regard to this issue and the importance of recognizing autism in children, this article explores whether people with autism can be screened by analyzing the way they play an emotion-based video game.

Social skills are one of the most important abilities people learn from the early months of their lives. These skills are extremely critical and affect all aspect of our lives from childhood to adulthood because they develop other skills and help us to adapt. Autistic children mostly show serious shortcomings in their social skills from the very first months of their lives. One of these social skills is the perception of emotion. We are unable to understand any emotions unless we can distinguish them from one another and identify them in ourselves or in others. Therefore, the first skill that people are required to learn is "the detection and recognition of emotions." In the next step, they must be able to express their inner emotions on their face, or in other words, the ability to "express emotions". Ultimately, someone who has these two abilities will need to have the ability to "regulate the emotions". Emotion regulation is the process by which individuals can influence what emotions they feel, when and how to experience and express them. In this research, a video game designed and used for autism screening because it can be easily accessible to everyone.

II. Game Design

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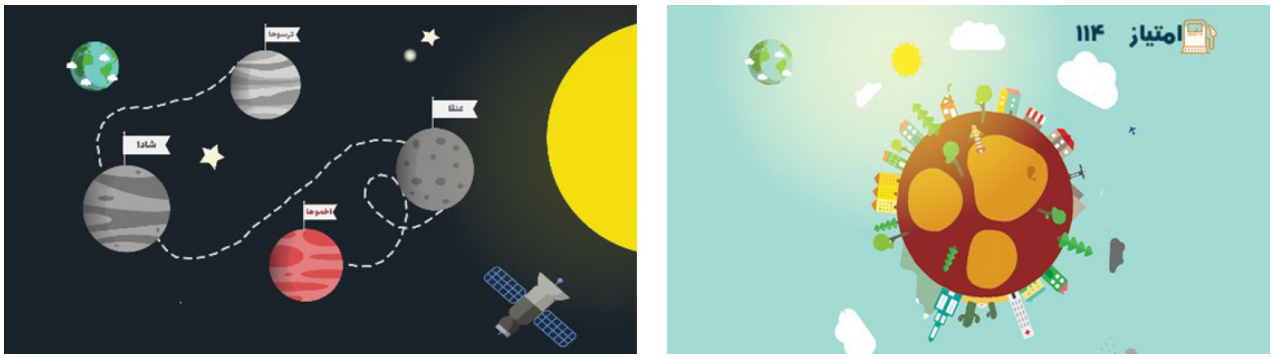

FIGURE I. (A) FOUR PLANETS IN THE GAME (B) A PLANET IN THE GAME

FIGURE II. EMOTION DETECTION GAMES (A) THE SEA GAME (B) THE SNOWBALL GAME

The "EmoGalaxy" game takes place in a system composed of four planets. Each planet refers to one of the main emotions. These four feelings are happiness, sadness, fear, and anger. There are a number of buildings on each planet, clicking on each one will open a game. For example, the "frowning" planet is related to anger, and the games that are on the planet are related to anger.

To enter each game, it is necessary for the children to show the feelings of that planet in their face. For example, to enter a game on the frowning planet, it is necessary for the child to express an angry face. Each game is also aimed at regulating, detecting or expressing emotions. The user receives a score after finishing each game. When the user points reach a limit, they actually collect fuel for their spacecraft and are able to go to another planet.

EmoGalaxy runs on Android smartphones and tablets with operating systems above 4.4 and iPhone 5 and above. The game is two-dimensional and is produced by Unity 5, one of the most powerful tools for game development and design. The game is designed to recognize, regulate, and express emotions. In the following, the training method for each of these three items in the game is described.

A. *Emotion Detection*

It is necessary for children to become familiar with an emotion before they can recognize it in their faces. Therefore, in two parts of the game section, each emotion is introduced through different faces, cartoon masks, and animated characters. When the players become familiar with feelings, in two games of "sea" and "snowball" they are asked to identify a person's expression in a number of photos.

1) *Sea*

There are 9 fish in the game and there is a person's image on each one. At the beginning of the game, the user is asked to select the fish that shows a specific feeling. If the users choose the right fish, they will be rewarded with a golden fish but if they choose incorrectly, the fish dies.

2) *Snowball*

Similar to the previous game, it aims to teach detecting the emotions. In this game, five snowmen with different feelings are displayed to the user and the players are asked to choose a snowman which has a certain feeling. If the answer is correct, a snowball will be thrown towards the snowman and if it is wrong, it will be thrown towards the user.

B. *Emotion Expression*

This part measures the players' ability to express their feelings on their face through the artificial intelligence implemented in the game. In fact, the game captures the users' images from the front-facing camera and recognizes the number of emotions on their face.

1) *The face scan*

Before entering any game, the users must show the emotion related to that planet on their face. For example, if users are on the frowning planet and click a game, they must show anger on their face before joining the game, and if they do it correctly, they will enter the game. If users cannot express that feeling on their face for more than 5 seconds, they are asked to choose that feeling from the other two options in order to continue playing.

C. *Emotion regulation*

Storytelling and Q&A have been used to teach emotion regulation skills. For example, in the "cloudy snakes and

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FIGURE III. FACE SCANNER SECTION, IN THE BEGINNING OF EACH GAME

ladders" section, the user moves forward if they can answer the questions about the emotion regulation correctly. In the "Story of a Feeling" section, the game also tries to explain the emotion regulation skills to the user by telling a story.

III. METHODS

A. Participants

Participants of the present study included 22 normal and 23 children with autism spectrum disorders. Normal children included non-governmental elementary school students from Tehran. The sample consisted of 22 children selected through available sampling method from two informal girls in Tehran. Their average age was 8 years.

Six young adults with low functioning ASD, ages 10-13 year (M=11) and 17 high functioning ASD people with age 6-14 year (M=12) participated in this study. All participants were selected from three autism center and had established clinical diagnoses of ASD.

B. Procedures

To assess the screening capabilities of the game, the game was presented to the selected people. Each of these children played the games related to the expression and detection of emotions. Since only children who were more powerful in detecting emotions entered emotional games, emotional regulation data was removed from screening and only the data of the sea and snowball (emotion detection) and face scans (emotion expression) were screened. The most important data is the player's score per game, which is calculated for each game using the following formula:

$$score = \frac{number\ of\ true\ answers}{number\ of\ moves}$$

Using data from these games and classification algorithms Screening is done.

C. Classification algorithms

When categorizing data, there are many methods in machine learning to choose from. The best option can vary depending on the question and the nature of data. The methods chosen here were support vector machine (SVM), decision tree, KNN, and logistic regression. Support vector machine is a popular method and has been used in many similar studies. Decision tree provides an interpretable presentation of data. The reason behind choosing these two methods is that they are robust to outliers. Usually, young children display a various range of behaviors; therefore data in different categories can seem quite similar. This is the case for children with autism; they display unspecified behaviors with quite some fluctuation; such that their performances can differ throughout the days. It is important to note that using specifically designed game for diagnosing autism in children is a method that has not been used before.

IV. RESULTS

The average score per user of these three games has been shown in charts.

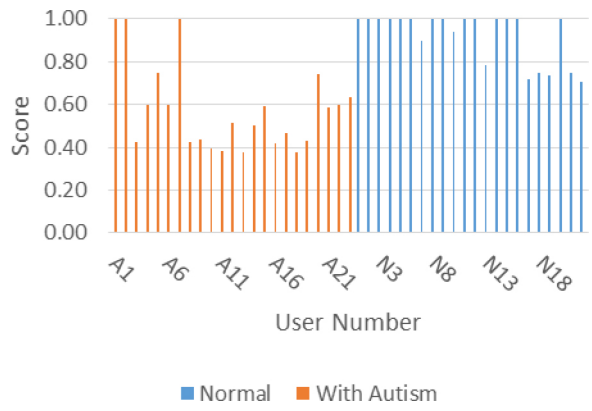

FIGURE IV. AVERAGE SCORES OF EACH PLAYER IN THE SEA GAME

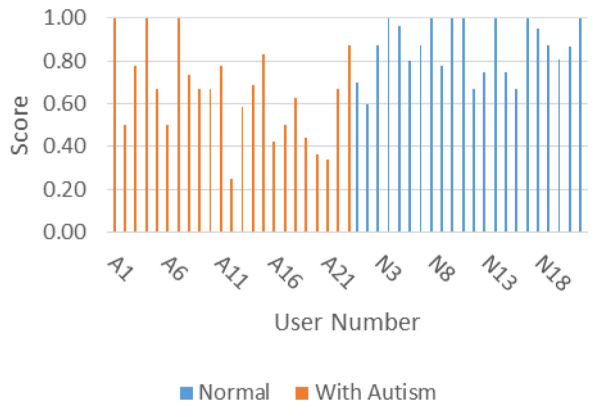

FIGURE V. AVERAGE SCORES OF EACH PLAYER IN THE SNOW BALL GAME

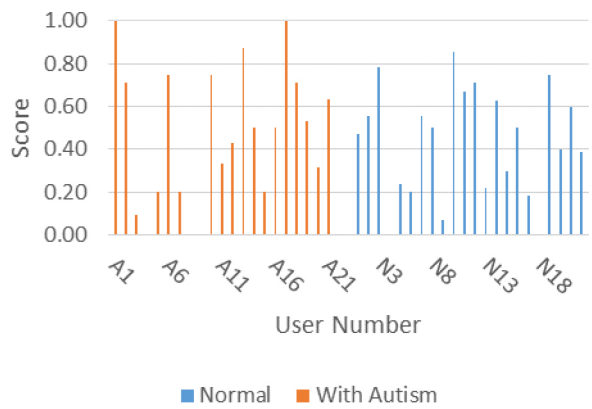

FIGURE VI. AVERAGE SCORES OF EACH PLAYER IN THE FACE SCANNER

It now needs to be checked whether "the data of the three selected games are suitable for screening?" Or "There is a difference between the data of normal and with autism

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children?". To analyze this issue, analysis of variance (ANOVA) is generally used. Before data can be used for this analysis, it must be ensured that the data has the required conditions for this analysis. So the normal distribution of data has been investigated using the Kolmogorov-Smirnov test.

According to the results of Table 1_and considering the significance level or α equal to 0.05, the null hypothesis is rejected for all three data, that is, none of these data has come from normal distribution and cannot be used for ANOVA. For this reason, the Kruskal-Wallis test has been used to no need for data to be normal.

TABLE I. KOLMOGOROV-SMIRNOV TEST RESULTS

Data	P-value	Result
Score of Sea Game	$P < 0.05$	Rejection of the null hypothesis
Score of Snowball Game	$P < 0.05$	Rejection of the null hypothesis
Score of Face Scanner	$P < 0.05$	Rejection of the null hypothesis

Table 2 shows the results of the Kruskal-Wallis test for snowball scores. Given the very low p-value (p=0/007) and $\alpha = 0.05$, the zero hypothesis is rejected. So, there are a difference between score of snowball for individuals with autism and normal children. This means that can use the game's data to screening.

TABLE II. KRUSKAL-WALLIS TEST FOR SNOWBALL SCORES

Source of Changes	SS	DF	MS	Chi-squared	P-value
Group	1961.17	1	1961.17	11.5	0.0007
Error	5540.83	43	128.86		
Total	7502	44			

According to Fig. 7, which shows the dispersion of snowball data for two groups of children, the median range of game scores varies between two groups of children. Therefore, this criterion can be used to separate two groups of children.

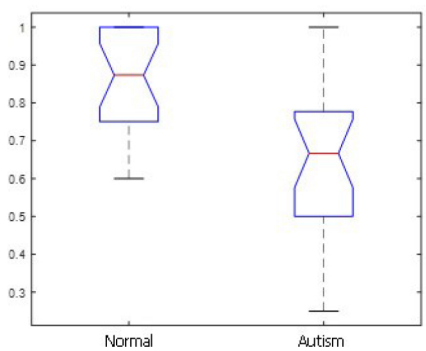

FIGURE VII. DISTRIBUTION OF SNOW BALL GAME SCORES FOR TWO GROUPS OF NORMAL AND CHILDREN WITH AUTISM

Table 3 shows the results of the Kruskal-Wallis test for sea scores. Given the very low p-value (p<0.05), the zero hypothesis is rejected. So, there are a difference between score of sea game for individuals with autism and normal children. This means that can use the game's data to screening. the median range of game scores varies between two groups of children(Fig. 8).

TABLE III. KRUSKAL-WALLIS TEST FOR SEA SCORES

Source of Changes	SS	DF	MS	Chi-squared	P-value
Group	3575.12	1	3575.12	21.91	0.00002
Error	3603.88	43	83.81		
Total	7179	44			

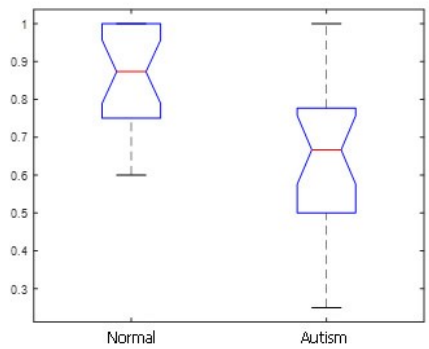

FIGURE VIII. DISTRIBUTION OF SEA GAME SCORES FOR TWO GROUPS OF NORMAL AND CHILDREN WITH AUTISM

the results of the variance analysis of face scanning show the p=0.83 and this value is higher than the determined α, so the game's data of people in these two groups do not differ significantly from this point of view (Table 4). This maybe because of the dependability of the results of this part of the game on the quality of the camera and the ambient. The problem that caused the results of normal children's play is not significantly different from children with autism.

TABLE IV. KRUSKAL-WALLIS TEST FOR FACE SCANNING

Source of Changes	SS	DF	MS	Chi-squared	P-value
Group	8.03	1	8.03	0.05	0.83
Error	7538.97	43	7538.97		
Total	7547	44			

A. *Classification*

The features selected for the implementation of classification algorithms are sea and snowball game's score that are related to the emotion detection. The number of folds in all of these classification algorithms was eight to prevent over fit.

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As shown in Table 5, the best result is for the classification using support vector machine (SVM) with Gaussian kernel.

TABLE V. RESULTS OF CLASSIFICATION WITH SUGGESTED ALGORITHMS

Gaussian SVM	Linear SVM	Logistic Regression	Decision Tree	KNN
93.3%	89.9%	88.9%	91.1%	91.1%

TABLE VI. CONFUSION MATRIX OF SELECTED ALGORITHM CLASSIFICATION

		Predicted Class	
		Normal	*Autism*
Actual Class	*Normal*	21	1
	Autism	2	21

In Fig. 9(a), data separation using the support vector machine with a Gaussian kernel function is shown. In Fig. 9(b) The same separation is plotted using a linear support vector machine. In Gaussian SVM the number of support vectors is high and the line segment does not have a definite shape. But in the second case with 3% less precision, the two classes are linearly separated. The advantage of this separator is that it is easier to implement and possibly less over fitted.

V. DISCUSSION

Serious games are a new and remarkable method for treating emotional problems in children. The objective is to use game designing tools in order to create a game that is not only entertaining, but it also helps children with their emotional difficulties. Given the growing use of smartphones in the last decade, games can be considered a suitable, vastly available and low-cost tool for treating children.

In this study, we hypothesized that screening with the Emogalaxy game could distinguish children with autism from healthy ones. Four algorithms of support vector machine, decision tree, logistic regression, and KNN were applied to data and achieved scores in the games. Some of the subjects were not adequately familiar with emotions and could not properly play the emotion regulation games; therefore data concerning this section was omitted. The results show that the best results are achieved through support vector machine with emotion recognition games data. This could be due to the dependence of emotion expression games to the participant's location and the qnality of the phone camera. Using this method, the screening has an accuracy of approximately 93%.

It is suggested that these games be tested with various samples and with different psychological disorders. Feedbacks about the way each group of children plays the games can help in developing and improving the game. According to feedback received from children's parent up to this point, children with autism need a simple interface to learn, without any additional audio and visual stimulus, while healthy children prefer more complex games with added audio stimulus and visual effects. Consequently, to make the game suitable for everyone, it should be developed separately for each unique group of children. Fortunately, the game strncture makes this a straightforward thing to do.

During this study, the parents provided feedback concerning the content and design of the game, which will be taken into account in the future versions. For instance, providing an option for playing in groups will able children to play together with their parents or siblings, which could have a good impact on the child's learning.

Comparing participants in different emotions could lead to some interesting results. For instance, how do normal and autistic children differ in their emotions of happiness and sadness? How do positive emotions in individuals with autism, such as happiness, differ from negative ones like fear and sadness? Due to limited time and the siuall sample in this study, such comparisons were not possible. However, future research with adeqnate data can answer these questions.

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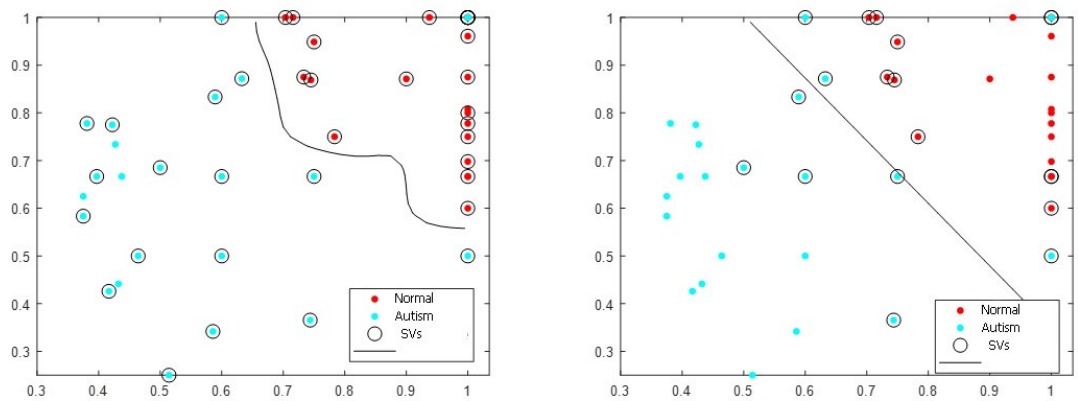

FIGURE IX. CLASSIFICATION DATA (A) USING GAUSSIAN KERNEL (B) USING LINEAR KERNEL

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